

Task for BD07 (2025 2026)

Exercises

1. Create a database called zoo.

USE zoo

2. Insert a collection called animals.

```
db.createCollection("animals")
```

show collections

```
C:\Users\sergi>mongosh "mongodb+srv://cluster0.sewaqxw.mongodb.net/" --apiVersion 1 --username
ssword 0ZD9cCUhuENchpBA
Current Mongosh Log ID: 69f77db9a44304b251abc113
Connecting to:      mongodb+srv://<credentials>@cluster0.sewaqxw.mongodb.net/?appName=mong
Using MongoDB:      8.0.21 (API Version 1)
Using Mongosh:      2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically
m/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

Atlas atlas-4gvgw6-shard-0 [primary] test> use zoo
switched to db zoo
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.createCollection("animals")
{ ok: 1 }
Atlas atlas-4gvgw6-shard-0 [primary] zoo> show collectons
MongoshInvalidInputError: [COMMON-10001] 'collectons' is not a valid argument for "show".
Atlas atlas-4gvgw6-shard-0 [primary] zoo> show collections
animals
Atlas atlas-4gvgw6-shard-0 [primary] zoo> |
```

3. Confirm that the database and the collection have been successfully created.

show dbs

show collections

```

Atlas atlas-4gvgw6-shard-0 [primary] zoo> show collectons
MongoshInvalidInputError: [COMMON-10001] 'collectons' is not
Atlas atlas-4gvgw6-shard-0 [primary] zoo> show collections
animals
Atlas atlas-4gvgw6-shard-0 [primary] zoo> show dbs
sample_mflix  111.66 MiB
zoo            8.00 KiB
admin          360.00 KiB
local          4.63 GiB
Atlas atlas-4gvgw6-shard-0 [primary] zoo> show collections
animals
Atlas atlas-4gvgw6-shard-0 [primary] zoo> |

```

4. Insert the documents contained in the animals.json file.

```
db.animals.insertMany([
```

```

{
  "_id": 1,
  "name": "Shere Khan",
  "age": 10,
  "gender": "Male",
  "species": "Tiger",
  "habitat": "Rainforest",
  "birthplace": "Sundarbans, India",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 101, "nameCaretaker": "John Smith" }]
},
{
  "_id": 2,
  "name": "Rajah",
  "age": 8,

```

```
"gender": "Male",
"species": "Tiger",
"habitat": "Rainforest",
"birthplace": "Bandipur Tiger Reserve, India",
"parents": [
  { "name": "Shere Khan", "gender": "Male" },
  { "name": "Tala", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 102, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 3,
  "name": "Tala",
  "age": 7,
  "gender": "Female",
  "species": "Tiger",
  "habitat": "Rainforest",
  "birthplace": "Kaziranga National Park, India",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 103, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 4,
  "name": "Tigris",
  "age": 9,
```

```
"gender": "Female",
"species": "Tiger",
"habitat": "Rainforest",
"birthplace": "Chitwan National Park, Nepal",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 104, "nameCaretaker": "Olivia Harris" }]
},
{
  "_id": 5,
  "name": "Kumba",
  "age": 6,
  "gender": "Male",
  "species": "Tiger",
  "habitat": "Rainforest",
  "birthplace": "Periyar National Park, India",
  "parents": [
    { "name": "Shere Khan", "gender": "Male" },
    { "name": "Tigris", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 105, "nameCaretaker": "Sophia White" }]
},
{
  "_id": 6,
  "name": "Zara",
  "age": 5,
```

```
"gender": "Female",
"species": "Tiger",
"habitat": "Rainforest",
"birthplace": "Sundarbans, India",
"parents": [
  { "name": "Rajah", "gender": "Male" },
  { "name": "Tala", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 106, "nameCaretaker": "Nathan Green" }]
},
{
  "_id": 7,
  "name": "Kiran",
  "age": 4,
  "gender": "Male",
  "species": "Tiger",
  "habitat": "Rainforest",
  "birthplace": "Chitwan National Park, Nepal",
  "parents": [
    { "name": "Kumba", "gender": "Male" },
    { "name": "Tigris", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 107, "nameCaretaker": "Michael Brown" }]
},
{
  "_id": 8,
  "name": "Mufasa",
  "age": 10,
```

```
"gender": "Male",
"species": "Lion",
"habitat": "Savanna",
"birthplace": "Serengeti National Park, Tanzania",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 201, "nameCaretaker": "John Smith" }]
},
{
  "_id": 9,
  "name": "Sarabi",
  "age": 9,
  "gender": "Female",
  "species": "Lion",
  "habitat": "Savanna",
  "birthplace": "Mufasa's Pride Zoo",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 202, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 10,
  "name": "Simba",
  "age": 4,
```

```
"gender": "Male",
"species": "Lion",
"habitat": "Savanna",
"birthplace": "Mufasa's Pride Zoo",
"parents": [
  { "name": "Mufasa", "gender": "Male" },
  { "name": "Sarabi", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 203, "nameCaretaker": "Alice Green" }]
},
{
  "_id": 11,
  "name": "Nala",
  "age": 6,
  "gender": "Female",
  "species": "Lion",
  "habitat": "Savanna",
  "birthplace": "Nairobi Zoo, Kenya",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 204, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 12,
  "name": "Scar",
  "age": 12,
```

```
"gender": "Male",
"species": "Lion",
"habitat": "Savanna",
"birthplace": "Serengeti National Park, Tanzania",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 205, "nameCaretaker": "Michael Davis" }]
},
{
  "_id": 13,
  "name": "Dumbo",
  "age": 25,
  "gender": "Male",
  "species": "Elephant",
  "habitat": "Grasslands",
  "birthplace": "Serengeti National Park, Tanzania",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 301, "nameCaretaker": "John Smith" }]
},
{
  "_id": 14,
  "name": "Jumbo",
  "age": 30,
```



```
"gender": "Male",
"species": "Elephant",
"habitat": "Grasslands",
"birthplace": "Kruger National Park, South Africa",
"parents": [
  { "name": "Dumbo", "gender": "Male" },
  { "name": "Ella", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 302, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 15,
  "name": "Ella",
  "age": 20,
  "gender": "Female",
  "species": "Elephant",
  "habitat": "Grasslands",
  "birthplace": "Serengeti National Park, Tanzania",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 303, "nameCaretaker": "Alice Green" }]
},
{
  "_id": 16,
  "name": "Tantor",
  "age": 15,
```

```
"gender": "Male",
"species": "Elephant",
"habitat": "Grasslands",
"birthplace": "Kruger National Park, South Africa",
"parents": [
  { "name": "Dumbo", "gender": "Male" },
  { "name": "Ella", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 304, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 17,
  "name": "Balu",
  "age": 12,
  "gender": "Male",
  "species": "Bear",
  "habitat": "Mountain Forest",
  "birthplace": "Rocky Mountains, USA",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 401, "nameCaretaker": "Jake Adams" }]
},
{
  "_id": 18,
  "name": "Kenai",
  "age": 8,
```

```
"gender": "Female",
"species": "Bear",
"habitat": "Mountain Forest",
"birthplace": "Yellowstone National Park, USA",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 402, "nameCaretaker": "Sophia Green" }]
},
{
  "_id": 19,
  "name": "Koda",
  "age": 6,
  "gender": "Male",
  "species": "Bear",
  "habitat": "Mountain Forest",
  "birthplace": "Denali National Park, USA",
  "parents": [
    { "name": "Balu", "gender": "Male" },
    { "name": "Kenai", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 403, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 20,
  "name": "Yogi",
  "age": 10,
```

```
"gender": "Male",
"species": "Bear",
"habitat": "Mountain Forest",
"birthplace": "Glacier National Park, USA",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 404, "nameCaretaker": "Michael Davis" }]
},
{
  "_id": 21,
  "name": "Smokey",
  "age": 11,
  "gender": "Male",
  "species": "Bear",
  "habitat": "Mountain Forest",
  "birthplace": "Appalachian Mountains, USA",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 405, "nameCaretaker": "John Smith" }]
},
{
  "_id": 22,
  "name": "Bagheera",
  "age": 8,
```

```
"gender": "Male",
"species": "Panther",
"habitat": "Rainforest",
"birthplace": "Amazon Rainforest, Brazil",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 406, "nameCaretaker": "Alice Green" }]
},
{
  "_id": 23,
  "name": "Shadow",
  "age": 5,
  "gender": "Male",
  "species": "Panther",
  "habitat": "Rainforest",
  "birthplace": "Amazon Rainforest, Brazil",
  "parents": [
    { "name": "Bagheera", "gender": "Male" },
    { "name": "Luna", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 407, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 24,
  "name": "Luna",
  "age": 7,
```

```
"gender": "Female",
"species": "Panther",
"habitat": "Rainforest",
"birthplace": "Amazon Rainforest, Brazil",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 408, "nameCaretaker": "Nathan Green" }]
},
{
  "_id": 25,
  "name": "Fang",
  "age": 6,
  "gender": "Male",
  "species": "Wolf",
  "habitat": "Tundra",
  "birthplace": "Alaska, USA",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 409, "nameCaretaker": "Emily White" }]
},
{
  "_id": 26,
  "name": "Luna",
  "age": 5,
```

```
"gender": "Female",
"species": "Wolf",
"habitat": "Tundra",
"birthplace": "Yukon, Canada",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 410, "nameCaretaker": "Sophia Taylor" }]
},
{
  "_id": 27,
  "name": "Ghost",
  "age": 7,
  "gender": "Male",
  "species": "Wolf",
  "habitat": "Tundra",
  "birthplace": "Yellowstone National Park, USA",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 411, "nameCaretaker": "Michael Brown" }]
},
{
  "_id": 28,
  "name": "Shadow",
  "age": 6,
```

```
"gender": "Male",
"species": "Wolf",
"habitat": "Tundra",
"birthplace": "Alaska, USA",
"parents": [
  { "name": "Fang", "gender": "Male" },
  { "name": "Luna", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 412, "nameCaretaker": "John White" }]
},
{
  "_id": 29,
  "name": "Snow",
  "age": 5,
  "gender": "Female",
  "species": "Wolf",
  "habitat": "Tundra",
  "birthplace": "Denali National Park, USA",
  "parents": [
    { "name": "Ghost", "gender": "Male" },
    { "name": "Luna", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 413, "nameCaretaker": "Nathan Hall" }]
},
{
  "_id": 30,
  "name": "Blaze",
  "age": 8,
```



```
"gender": "Male",
"species": "Wolf",
"habitat": "Tundra",
"birthplace": "Yukon, Canada",
"parents": [
  { "name": "Fang", "gender": "Male" },
  { "name": "Luna", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 414, "nameCaretaker": "Sophia Green" }]
},
{
  "_id": 31,
  "name": "Storm",
  "age": 6,
  "gender": "Male",
  "species": "Wolf",
  "habitat": "Tundra",
  "birthplace": "Yellowstone National Park, USA",
  "parents": [
    { "name": "Ghost", "gender": "Male" },
    { "name": "Luna", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 415, "nameCaretaker": "Alice Brown" }]
}, {
  "_id": 32,
  "name": "Rattle",
  "age": 5,
  "gender": "Male",
```

```
"species": "Rattlesnake",
"habitat": "Desert",
"birthplace": "Arizona, USA",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 501, "nameCaretaker": "Jake Adams" }]
},
{
  "_id": 33,
  "name": "Freedom",
  "age": 6,
  "gender": "Male",
  "species": "Eagle",
  "habitat": "Mountains",
  "birthplace": "Rocky Mountains, USA",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 502, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 34,
  "name": "Liberty",
  "age": 4,
  "gender": "Female",
```

```
"species": "Eagle",
"habitat": "Mountains",
"birthplace": "Yellowstone National Park, USA",
"parents": [
  { "name": "Freedom", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 503, "nameCaretaker": "John White" }]
},
{
  "_id": 35,
  "name": "Rafiki",
  "age": 12,
  "gender": "Male",
  "species": "Mandrill",
  "habitat": "Rainforest",
  "birthplace": "Gabon",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 504, "nameCaretaker": "Alice Green" }]
},
{
  "_id": 36,
  "name": "Kibo",
  "age": 10,
  "gender": "Male",
```

```
"species": "Mandrill",
"habitat": "Rainforest",
"birthplace": "Congo Basin",
"parents": [
  { "name": "Rafiki", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 505, "nameCaretaker": "Sophia Taylor" }]
},
{
  "_id": 37,
  "name": "Zuri",
  "age": 8,
  "gender": "Female",
  "species": "Mandrill",
  "habitat": "Rainforest",
  "birthplace": "Congo Basin",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 506, "nameCaretaker": "Mark Brown" }]
},
{
  "_id": 38,
  "name": "Jambo",
  "age": 6,
  "gender": "Male",
```

```
"species": "Mandrill",
"habitat": "Rainforest",
"birthplace": "Gabon",
"parents": [
  { "name": "Rafiki", "gender": "Male" },
  { "name": "Zuri", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 507, "nameCaretaker": "Nathan Green" }]
},
{
  "_id": 39,
  "name": "Nia",
  "age": 7,
  "gender": "Female",
  "species": "Mandrill",
  "habitat": "Rainforest",
  "birthplace": "Congo Basin",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 508, "nameCaretaker": "Olivia Davis" }]
},
{
  "_id": 40,
  "name": "Tamu",
  "age": 5,
  "gender": "Male",
```

```
"species": "Mandrill",
"habitat": "Rainforest",
"birthplace": "Gabon",
"parents": [
  { "name": "Rafiki", "gender": "Male" },
  { "name": "Zuri", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 509, "nameCaretaker": "Jake Adams" }]
},
{
  "_id": 41,
  "name": "Chuma",
  "age": 4,
  "gender": "Female",
  "species": "Mandrill",
  "habitat": "Rainforest",
  "birthplace": "Congo Basin",
  "parents": [
    { "name": "Jambo", "gender": "Male" },
    { "name": "Nia", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 510, "nameCaretaker": "Sophia White" }]
},
{
  "_id": 42,
  "name": "Mia",
  "age": 5,
  "gender": "Female",
```

```
"species": "Spider Monkey",
"habitat": "Rainforest",
"birthplace": "Amazon Rainforest, Brazil",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 511, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 43,
  "name": "Lola",
  "age": 6,
  "gender": "Female",
  "species": "Spider Monkey",
  "habitat": "Rainforest",
  "birthplace": "Amazon Rainforest, Brazil",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 512, "nameCaretaker": "Emily Green" }]
},
{
  "_id": 44,
  "name": "Coco",
  "age": 7,
  "gender": "Male",
```

```
"species": "Spider Monkey",
"habitat": "Rainforest",
"birthplace": "Amazon Rainforest, Brazil",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 513, "nameCaretaker": "Nathan Green" }]
},
{
  "_id": 45,
  "name": "Max",
  "age": 6,
  "gender": "Male",
  "species": "Coyote",
  "habitat": "Desert",
  "birthplace": "Nevada, USA",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 514, "nameCaretaker": "Sophia Taylor" }]
},
{
  "_id": 46,
  "name": "Luna",
  "age": 5,
  "gender": "Female",
```



```
"species": "Coyote",
"habitat": "Desert",
"birthplace": "Texas, USA",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 515, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 47,
  "name": "Shadow",
  "age": 6,
  "gender": "Male",
  "species": "Coyote",
  "habitat": "Desert",
  "birthplace": "Nevada, USA",
  "parents": [
    { "name": "Max", "gender": "Male" },
    { "name": "Luna", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 516, "nameCaretaker": "John White" }]
},
{
  "_id": 48,
  "name": "Maya",
  "age": 5,
  "gender": "Female",
```

```
"species": "Coyote",
"habitat": "Desert",
"birthplace": "Texas, USA",
"parents": [
  { "name": "Max", "gender": "Male" },
  { "name": "Luna", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 517, "nameCaretaker": "Sophia Green" }]
},
{
  "_id": 49,
  "name": "Blaze",
  "age": 7,
  "gender": "Male",
  "species": "Coyote",
  "habitat": "Desert",
  "birthplace": "Nevada, USA",
  "parents": [
    { "name": "Max", "gender": "Male" },
    { "name": "Luna", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 518, "nameCaretaker": "Emily White" }]
},
{
  "_id": 50,
  "name": "Storm",
  "age": 6,
  "gender": "Male",
```

```
"species": "Coyote",
"habitat": "Desert",
"birthplace": "Texas, USA",
"parents": [
  { "name": "Max", "gender": "Male" },
  { "name": "Luna", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 519, "nameCaretaker": "Nathan Hall" }]
},
{
  "_id": 51,
  "name": "Shere Khan",
  "age": 10,
  "gender": "Male",
  "species": "Tiger",
  "habitat": "Rainforest",
  "birthplace": "Sundarbans, India",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 520, "nameCaretaker": "Michael Brown" }]
},
{
  "_id": 52,
  "name": "Rajah",
  "age": 8,
  "gender": "Male",
```

```
"species": "Tiger",
"habitat": "Rainforest",
"birthplace": "Bandipur Tiger Reserve, India",
"parents": [
  { "name": "Shere Khan", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 521, "nameCaretaker": "Emily Green" }]
},
{
  "_id": 53,
  "name": "Tala",
  "age": 7,
  "gender": "Female",
  "species": "Tiger",
  "habitat": "Rainforest",
  "birthplace": "Kaziranga National Park, India",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 522, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 54,
  "name": "Buffy",
  "age": 8,
  "gender": "Male",
```

```
"species": "Buffalo",
"habitat": "Grasslands",
"birthplace": "Serengeti National Park, Tanzania",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 523, "nameCaretaker": "Alice White" }]
},
{
  "_id": 55,
  "name": "Bison",
  "age": 9,
  "gender": "Male",
  "species": "Buffalo",
  "habitat": "Grasslands",
  "birthplace": "Serengeti National Park, Tanzania",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 524, "nameCaretaker": "Nathan Hall" }]
},
{
  "_id": 56,
  "name": "Grazor",
  "age": 7,
  "gender": "Female",
```

```
"species": "Buffalo",
"habitat": "Grasslands",
"birthplace": "Serengeti National Park, Tanzania",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 525, "nameCaretaker": "Jake Adams" }]
},
{
  "_id": 57,
  "name": "Joey",
  "age": 5,
  "gender": "Male",
  "species": "Kangaroo",
  "habitat": "Grasslands",
  "birthplace": "Australia",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 601, "nameCaretaker": "Alice Brown" }]
},
{
  "_id": 58,
  "name": "Ruby",
  "age": 4,
  "gender": "Female",
```

```
"species": "Kangaroo",
"habitat": "Grasslands",
"birthplace": "Australia",
"parents": [
  { "name": "Joey", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 602, "nameCaretaker": "Emily Green" }]
},
{
  "_id": 59,
  "name": "Skip",
  "age": 6,
  "gender": "Male",
  "species": "Kangaroo",
  "habitat": "Grasslands",
  "birthplace": "Australia",
  "parents": [
    { "name": "Joey", "gender": "Male" },
    { "name": "Ruby", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 603, "nameCaretaker": "Jake Adams" }]
},
{
  "_id": 60,
  "name": "Dune",
  "age": 7,
  "gender": "Male",
```

```
"species": "Dromedary",
"habitat": "Desert",
"birthplace": "Sahara, Africa",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 604, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 61,
  "name": "Sandy",
  "age": 5,
  "gender": "Female",
  "species": "Dromedary",
  "habitat": "Desert",
  "birthplace": "Sahara, Africa",
  "parents": [
    { "name": "Dune", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 605, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 62,
  "name": "Hump",
  "age": 6,
  "gender": "Male",
```



```
"species": "Dromedary",
"habitat": "Desert",
"birthplace": "Sahara, Africa",
"parents": [
  { "name": "Dune", "gender": "Male" },
  { "name": "Sandy", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 606, "nameCaretaker": "Nathan Green" }]
},
{
  "_id": 63,
  "name": "Hippo",
  "age": 15,
  "gender": "Male",
  "species": "Hippopotamus",
  "habitat": "River",
  "birthplace": "Nile River, Africa",
  "parents": [
    { "name": "Unknown", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 607, "nameCaretaker": "Alice Green" }]
},
{
  "_id": 64,
  "name": "Muddy",
  "age": 12,
  "gender": "Female",
```

```
"species": "Hippopotamus",
"habitat": "River",
"birthplace": "Nile River, Africa",
"parents": [
  { "name": "Hippo", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 608, "nameCaretaker": "Mark Brown" }]
},
{
  "_id": 65,
  "name": "Splash",
  "age": 10,
  "gender": "Male",
  "species": "Hippopotamus",
  "habitat": "River",
  "birthplace": "Nile River, Africa",
  "parents": [
    { "name": "Hippo", "gender": "Male" },
    { "name": "Muddy", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 609, "nameCaretaker": "Sophia Taylor" }]
},
{
  "_id": 66,
  "name": "Spot",
  "age": 12,
  "gender": "Male",
```

```
"species": "Giraffe",
"habitat": "Savanna",
"birthplace": "Serengeti, Tanzania",
"parents": [
  { "name": "Unknown", "gender": "Male" },
  { "name": "Unknown", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 610, "nameCaretaker": "Michael Davis" }]
},
{
  "_id": 67,
  "name": "Grace",
  "age": 11,
  "gender": "Female",
  "species": "Giraffe",
  "habitat": "Savanna",
  "birthplace": "Serengeti, Tanzania",
  "parents": [
    { "name": "Spot", "gender": "Male" },
    { "name": "Unknown", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 611, "nameCaretaker": "Sophia Green" }]
},
{
  "_id": 68,
  "name": "Tallie",
  "age": 9,
  "gender": "Female",
```

```
"species": "Giraffe",
"habitat": "Savanna",
"birthplace": "Serengeti, Tanzania",
"parents": [
  { "name": "Spot", "gender": "Male" },
  { "name": "Grace", "gender": "Female" }
],
"caretakers": [{ "idCaretaker": 612, "nameCaretaker": "Mark Taylor" }]
},
{
  "_id": 69,
  "name": "Neckie",
  "age": 8,
  "gender": "Male",
  "species": "Giraffe",
  "habitat": "Savanna",
  "birthplace": "Serengeti, Tanzania",
  "parents": [
    { "name": "Spot", "gender": "Male" },
    { "name": "Grace", "gender": "Female" }
  ],
  "caretakers": [{ "idCaretaker": 613, "nameCaretaker": "Emily Brown" }]
},
{
  "_id": 70,
  "name": "Horn",
  "age": 7,
  "gender": "Male",
```

```

    "species": "Rhinoceros",
    "habitat": "Grasslands",
    "birthplace": "Kenya",
    "parents": [
      { "name": "Unknown", "gender": "Male" },
      { "name": "Unknown", "gender": "Female" }
    ],
    "caretakers": [{ "idCaretaker": 614, "nameCaretaker": "Nathan Hall" }]
  }
})

```

```

'44': 45,
'45': 46,
'46': 47,
'47': 48,
'48': 49,
'49': 50,
'50': 51,
'51': 52,
'52': 53,
'53': 54,
'54': 55,
'55': 56,
'56': 57,
'57': 58,
'58': 59,
'59': 60,
'60': 61,
'61': 62,
'62': 63,
'63': 64,
'64': 65,
'65': 66,
'66': 67,
'67': 68,
'68': 69,
'69': 70
}
}
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.countDocuments()
70

```

5- Write a MongoDB query to find animals that have Rainforest as their habitat and are attended by a caretaker named Emily Brown.

```
db.animals.find({ habitat: "Rainforest" }).count()
```

```
db.animals.find({ "caretakers.nameCaretaker": "Emily Brown" }).count()
```

```
db.animals.find({  
  | habitat: "Rainforest",  
  | "caretakers.nameCaretaker": "Emily Brown"  
  | }).count()
```

```
]
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find({ habitat: "Rainforest" }).count()
23
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find({ "caretakers.nameCaretaker": "Emily Brown" }).count()
8
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find({
  |   habitat: "Rainforest",
  |   "caretakers.nameCaretaker": "Emily Brown"
  | }).count()
1
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find({
  |   habitat: "Rainforest",
  |   "caretakers.nameCaretaker": "Emily Brown"
  | })
[
  {
    _id: 2,
    name: 'Rajah',
    age: 8,
    gender: 'Male',
    species: 'Tiger',
    habitat: 'Rainforest',
    birthplace: 'Bandipur Tiger Reserve, India',
    parents: [
      { name: 'Shere Khan', gender: 'Male' },
      { name: 'Tala', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 102, nameCaretaker: 'Emily Brown' } ]
  }
]
Atlas atlas-4gvgw6-shard-0 [primary] zoo>
```

6- Write a MongoDB query to display only the name, species, and habitat of all animals sorted by their species in ascending order.

```
db.animals.find(  
  {},  
  { name: 1, species: 1, habitat: 1, _id: 0 }  
).sort({ species: 1 })
```

```

Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find(
|   {},
|   { name: 1, species: 1, habitat: 1, _id: 0 }
| ).sort({ species: 1 })
[
  { name: 'Balu', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Kenai', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Koda', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Yogi', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Smokey', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Bison', species: 'Buffalo', habitat: 'Grasslands' },
  { name: 'Grazor', species: 'Buffalo', habitat: 'Grasslands' },
  { name: 'Buffy', species: 'Buffalo', habitat: 'Grasslands' },
  { name: 'Max', species: 'Coyote', habitat: 'Desert' },
  { name: 'Luna', species: 'Coyote', habitat: 'Desert' },
  { name: 'Shadow', species: 'Coyote', habitat: 'Desert' },
  { name: 'Maya', species: 'Coyote', habitat: 'Desert' },
  { name: 'Blaze', species: 'Coyote', habitat: 'Desert' },
  { name: 'Storm', species: 'Coyote', habitat: 'Desert' },
  { name: 'Hump', species: 'Dromedary', habitat: 'Desert' },
  { name: 'Dune', species: 'Dromedary', habitat: 'Desert' },
  { name: 'Sandy', species: 'Dromedary', habitat: 'Desert' },
  { name: 'Liberty', species: 'Eagle', habitat: 'Mountains' },
  { name: 'Freedom', species: 'Eagle', habitat: 'Mountains' },
  { name: 'Ella', species: 'Elephant', habitat: 'Grasslands' }
]
Type "it" for more
Atlas atlas-4gvgw6-shard-0 [primary] zoo>

```

7. Write a MongoDB query to find animals with the species Tiger and display only their name and age, sorted by age in descending order.

```

db.animals.find(
  { species: "Tiger" },
  { name: 1, age: 1, _id: 0 }
).sort({ age: -1 })

```

```
Type "it" for more
Atlas atlas-4gv6w6-shard-0 [primary] zoo> db.animals.find(
|   { species: "Tiger" },
|   { name: 1, age: 1, _id: 0 }
| ).sort({ age: -1 })
[
  { name: 'Shere Khan', age: 10 },
  { name: 'Shere Khan', age: 10 },
  { name: 'Tigris', age: 9 },
  { name: 'Rajah', age: 8 },
  { name: 'Rajah', age: 8 },
  { name: 'Tala', age: 7 },
  { name: 'Tala', age: 7 },
  { name: 'Kumba', age: 6 },
  { name: 'Zara', age: 5 },
  { name: 'Kiran', age: 4 }
]
Atlas atlas-4gv6w6-shard-0 [primary] zoo>
```

8. Write a MongoDB query to insert the next animal into the animals collection: name: "Leo", age: 7, gender: "Male", species: "Lion", habitat: "Savanna", birthplace: "Masai Mara, Kenya", parents: "Unknown", gender: "Male", "Unknown", gender: "Female" caretakers: id: 501, name: "Jake Adams"]]

```
db.animals.insertOne({
  name: "Leo",
  age: 7,
  gender: "Male",
  species: "Lion",
  habitat: "Savanna",
  birthplace: "Masai Mara, Kenya",
  parents: [
    { name: "Unknown", gender: "Male" },
    { name: "Unknown", gender: "Female" }
  ],
  caretakers: [
```



```

    { idCaretaker: 501, nameCaretaker: "Jake Adams" }
  ]
})

```

```
db.animals.find({ name: "Leo" })
```

```

Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.insertOne({
  name: "Leo",
  age: 7,
  gender: "Male",
  species: "Lion",
  habitat: "Savanna",
  birthplace: "Masai Mara, Kenya",
  parents: [
    { name: "Unknown", gender: "Male" },
    { name: "Unknown", gender: "Female" }
  ],
  caretakers: [
    { idCaretaker: 501, nameCaretaker: "Jake Adams" }
  ]
})
{
  acknowledged: true,
  insertedId: ObjectId('69f789fba44304b251abc114')
}
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find({ name: "Leo" })
[
  {
    _id: ObjectId('69f789fba44304b251abc114'),
    name: 'Leo',
    age: 7,
    gender: 'Male',
    species: 'Lion',
    habitat: 'Savanna',
    birthplace: 'Masai Mara, Kenya',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 501, nameCaretaker: 'Jake Adams' } ]
  }
]
Atlas atlas-4gvgw6-shard-0 [primary] zoo>

```

9. Write a MongoDB query to update the age (to 8) of the animal you just inserted using the \$set operator

```

db.animals.updateOne(
  { name: "Leo" },
  { $set: { age: 8 } }
)

```

)

```
db.animals.find({ name: "Leo" })
```

```
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.updateOne(
|   { name: "Leo" },
|   { $set: { age: 8 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find({ name: "Leo" })
[
  {
    _id: ObjectId('69f789fba44304b251abc114'),
    name: 'Leo',
    age: 8,
    gender: 'Male',
    species: 'Lion',
    habitat: 'Savanna',
    birthplace: 'Masai Mara, Kenya',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 501, nameCaretaker: 'Jake Adams' } ]
  }
]
Atlas atlas-4gvgw6-shard-0 [primary] zoo>
```

10. Write a MongoDB query to rename the habitat field to environment for all animals in the animals collection

```
db.animals.updateMany(
  {},
  { $rename: { "habitat": "environment" } }
)
```

```
db.animals.findOne()
```

```

]
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.updateMany(
|   {},
|   { $rename: { "habitat": "environment" } }
| )
|
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 71,
  modifiedCount: 71,
  upsertedCount: 0
}
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.findOne()
{
  _id: 1,
  name: 'Shere Khan',
  age: 10,
  gender: 'Male',
  species: 'Tiger',
  birthplace: 'Sundarbans, India',
  parents: [
    { name: 'Unknown', gender: 'Male' },
    { name: 'Unknown', gender: 'Female' }
  ],
  caretakers: [ { idCaretaker: 101, nameCaretaker: 'John Smith' } ],
  environment: 'Rainforest'
}
Atlas atlas-4gvgw6-shard-0 [primary] zoo>

```

11- Write a MongoDB query using the aggregation framework to find animals aged less than 10 and sort them alphabetically by their name.

```

db.animals.aggregate([
  { $match: { age: { $lt: 10 } } },
  { $sort: { name: 1 } }
])

```

```

    },
    {
      _id: 57,
      name: 'Joey',
      age: 5,
      gender: 'Male',
      species: 'Kangaroo',
      birthplace: 'Australia',
      parents: [
        { name: 'Unknown', gender: 'Male' },
        { name: 'Unknown', gender: 'Female' }
      ],
      caretakers: [ { idCaretaker: 601, nameCaretaker: 'Alice Brown' } ],
      environment: 'Grasslands'
    },
    {
      _id: 18,
      name: 'Kenai',
      age: 8,
      gender: 'Female',
      species: 'Bear',
      birthplace: 'Yellowstone National Park, USA',
      parents: [
        { name: 'Unknown', gender: 'Male' },
        { name: 'Unknown', gender: 'Female' }
      ],
      caretakers: [ { idCaretaker: 402, nameCaretaker: 'Sophia Green' } ],
      environment: 'Mountain Forest'
    },
    {
      _id: 7,
      name: 'Kiran',
      age: 4,
      gender: 'Male',
      species: 'Tiger',
      birthplace: 'Chitwan National Park, Nepal',
      parents: [
        { name: 'Kumba', gender: 'Male' },
        { name: 'Tigris', gender: 'Female' }
      ],
      caretakers: [ { idCaretaker: 107, nameCaretaker: 'Michael Brown' } ],

```

12. Write a MongoDB query using the aggregation framework to group animals by their

species and count the number of animals in each species.

```

db.animals.aggregate([
  {
    $group: {
      _id: "$species",

```

```

    total: { $sum: 1 }
  }
}
])

```

```

}
]
Type "it" for more
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.aggregate([
  {
    $group: {
      _id: "$species",
      total: { $sum: 1 }
    }
  }
])
[
  { _id: 'Spider Monkey', total: 3 },
  { _id: 'Mandrill', total: 7 },
  { _id: 'Giraffe', total: 4 },
  { _id: 'Rattlesnake', total: 1 },
  { _id: 'Lion', total: 6 },
  { _id: 'Kangaroo', total: 3 },
  { _id: 'Buffalo', total: 3 },
  { _id: 'Coyote', total: 6 },
  { _id: 'Rhinoceros', total: 1 },
  { _id: 'Hippopotamus', total: 3 },
  { _id: 'Panther', total: 3 },
  { _id: 'Wolf', total: 7 },
  { _id: 'Tiger', total: 10 },
  { _id: 'Elephant', total: 4 },
  { _id: 'Bear', total: 5 },
  { _id: 'Eagle', total: 2 },
  { _id: 'Dromedary', total: 3 }
]
Atlas atlas-4gvgw6-shard-0 [primary] zoo>

```

13. Write a MongoDB query to create a new collection called Caretakers by extracting

all caretakers from the animals collection. Ensure that caretakers do not repeat and

include an array of the animals they attend to.

```

db.animals.aggregate([
  { $unwind: "$caretakers" },

```

```
{
  $group: {
    _id: "$caretakers.idCaretaker",
    nameCaretaker: { $first: "$caretakers.nameCaretaker" },
    animals: { $addToSet: "$name" }
  }
},
{
  $project: {
    _id: 0,
    idCaretaker: "$_id",
    nameCaretaker: 1,
    animals: 1
  }
},
{ $out: "Caretakers" }
])
```

show collections

db.Caretakers.find()

```
}  
{  
  _id: ObjectId('69f78c969d088d83871e3d9d'),  
  nameCaretaker: 'Alice Brown',  
  animals: [ 'Storm' ],  
  idCaretaker: 415  
},  
{  
  _id: ObjectId('69f78c969d088d83871e3d9e'),  
  nameCaretaker: 'Jake Adams',  
  animals: [ 'Rattle', 'Leo' ],  
  idCaretaker: 501  
},  
{  
  _id: ObjectId('69f78c969d088d83871e3d9f'),  
  nameCaretaker: 'John Smith',  
  animals: [ 'Smokey' ],  
  idCaretaker: 405  
}  
]  
Type "it" for more  
Atlas atlas-4gvgw6-shard-0 [primary] zoo> |
```

14. Write a MongoDB query to find the top 3 oldest animals.

```
db.animals.find().sort({ age: -1 }).limit(3)
```

```

]
Type "it" for more
Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find().sort({ age: -1 }).limit(3)
[
  {
    _id: 14,
    name: 'Jumbo',
    age: 30,
    gender: 'Male',
    species: 'Elephant',
    birthplace: 'Kruger National Park, South Africa',
    parents: [
      { name: 'Dumbo', gender: 'Male' },
      { name: 'Ella', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 302, nameCaretaker: 'Emily Brown' } ],
    environment: 'Grasslands'
  },
  {
    _id: 13,
    name: 'Dumbo',
    age: 25,
    gender: 'Male',
    species: 'Elephant',
    birthplace: 'Serengeti National Park, Tanzania',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 301, nameCaretaker: 'John Smith' } ],
    environment: 'Grasslands'
  },
  {
    _id: 15,
    name: 'Ella',
    age: 20,
    gender: 'Female',
    species: 'Elephant',
    birthplace: 'Serengeti National Park, Tanzania',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 303, nameCaretaker: 'Alice Green' } ],
    environment: 'Grasslands'
  }
]

```

15. Write a MongoDB query to display animals between the 6th and 12th entries in the collection based on their insertion order.

```
db.animals.find().skip(5).limit(7)
```



```

Atlas atlas-4gvgw6-shard-0 [primary] zoo> db.animals.find().skip(5).limit(7)
[
  {
    _id: 6,
    name: 'Zara',
    age: 5,
    gender: 'Female',
    species: 'Tiger',
    birthplace: 'Sundarbans, India',
    parents: [
      { name: 'Rajah', gender: 'Male' },
      { name: 'Tala', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 106, nameCaretaker: 'Nathan Green' } ],
    environment: 'Rainforest'
  },
  {
    _id: 7,
    name: 'Kiran',
    age: 4,
    gender: 'Male',
    species: 'Tiger',
    birthplace: 'Chitwan National Park, Nepal',
    parents: [
      { name: 'Kumba', gender: 'Male' },
      { name: 'Tigris', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 107, nameCaretaker: 'Michael Brown' } ],
    environment: 'Rainforest'
  },
  {
    _id: 8,
    name: 'Mufasa',
    age: 10,
    gender: 'Male',
    species: 'Lion',
    birthplace: 'Serengeti National Park, Tanzania',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 201, nameCaretaker: 'John Smith' } ],
    environment: 'Savanna'
  },
]

```

16. Write a MongoDB query using the aggregation framework to create a new collection called Habitats. This collection should group animals by their environment (formerly habitat) and include the following fields.

- The environment name.
- The total number of animals in that environment.
- An array of unique species in that environment, sorted alphabetically.
- An array of unique caretakers who attend animals in that environment.

```

db.animals.aggregate([
  { $unwind: "$caretakers" },
  {
    $group: {
      _id: "$environment",
      animals: { $addToSet: "$_id" },
      species: { $addToSet: "$species" },
      caretakers: { $addToSet: "$caretakers.nameCaretaker" }
    }
  },
  {
    $project: {
      _id: 0,
      environment: "$_id",
      totalAnimals: { $size: "$animals" },
      species: {
        $sortBy: { input: "$species", sortBy: 1 }
      },
      caretakers: 1
    }
  },
  { $out: "Habitats" }
])

```

```

db.Habitats.find()

```

```

    {
      _id: ObjectId('69f791629d088d83871e3e87'),
      caretakers: [
        'Emily Brown',
        'Michael Davis',
        'John Smith',
        'Sophia Green',
        'Jake Adams'
      ],
      environment: 'Mountain Forest',
      totalAnimals: 5,
      species: [ 'Bear' ]
    },
    {
      _id: ObjectId('69f791629d088d83871e3e88'),
      caretakers: [ 'Mark Brown', 'Sophia Taylor', 'Alice Green' ],
      environment: 'River',
      totalAnimals: 3,
      species: [ 'Hippopotamus' ]
    },
    {
      _id: ObjectId('69f791629d088d83871e3e89'),
      caretakers: [ 'John White', 'Emily Brown' ],
      environment: 'Mountains',
      totalAnimals: 2,
      species: [ 'Eagle' ]
    },
    {
      _id: ObjectId('69f791629d088d83871e3e8a'),
      caretakers: [
        'Emily Brown',
        'Michael Davis',
        'Sophia Green',
        'Jake Adams',
        'Alice Green',
        'John Smith',
        'Mark Taylor'
      ],
      environment: 'Savanna',
      totalAnimals: 10,
      species: [ 'Giraffe', 'Lion' ]
    }
  ]
}
Atlas atlas-4gvw6-shard-0 [primary] zoo>

```